The critical points of an example function.

```
clear
```

```
syms x y
```

```

```

```
f(x,y) =-2*x^4 + 4*x^2*exp(y) - 4*exp(4*y)
```

f(x, y) = $-2\,x^4+4\,\mathrm{e}^y\,x^2-4\,\mathrm{e}^{4\,y}$

```
fx = diff(f,x)
```

fx(x, y) = $8\,x\,\mathrm{e}^y-8\,x^3$

```
fy = diff(f,y)
```

fy(x, y) = $4\,x^2\,\mathrm{e}^y-16\,\mathrm{e}^{4\,y}$

```
fxy = diff(fx,y)
```

fxy(x, y) = $8\,x\,\mathrm{e}^y$

```
fxx = diff(fx,x)
```

fxx(x, y) = $8\,\mathrm{e}^y-24\,x^2$

```
fyy = diff(fy,y)
```

fyy(x, y) = $4\,x^2\,\mathrm{e}^y-64\,\mathrm{e}^{4\,y}$

```
sol = solve([fx(x,y)==0,fy(x,y)==0],[x,y],"Real",true)
```

```
sol =
struct with fields:
    x: [2×1 sym]
    y: [2×1 sym]
```

```
sol.x
```

```
ans =
```

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{pmatrix}$$

```
sol.y
```

```
ans =
```

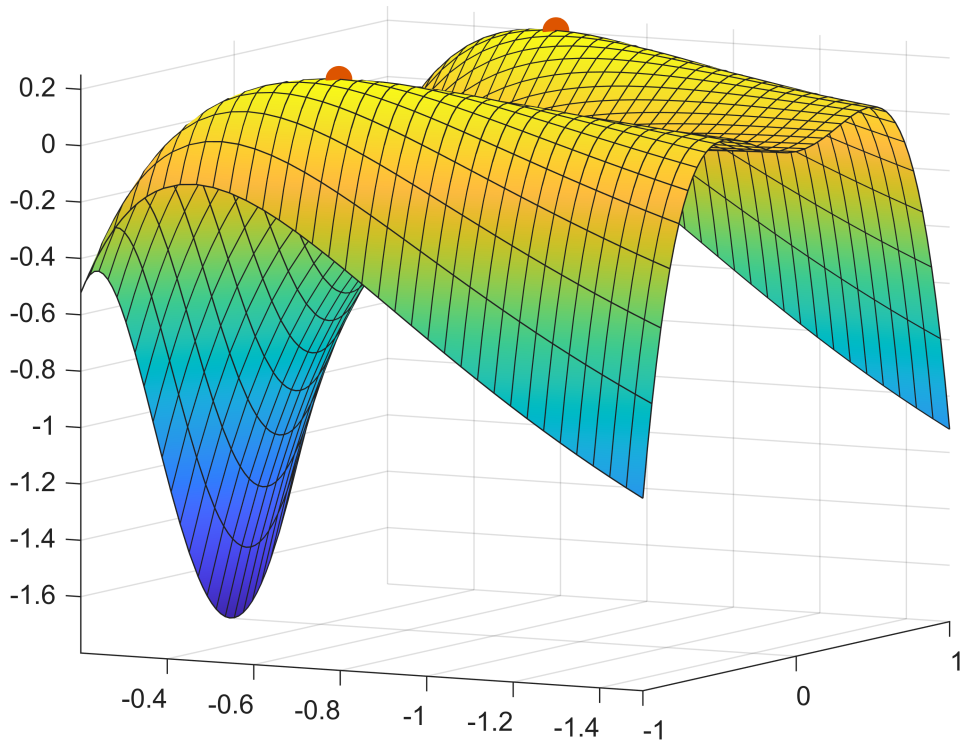
$$\begin{pmatrix} -\log(2) \\ -\log(2) \end{pmatrix}$$

```
for i=1:2
    disp([sol.x(i),sol.y(i)])
end
```

$$\begin{pmatrix} -\frac{\sqrt{2}}{2} & -\log(2) \end{pmatrix}$$

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\log(2) \end{pmatrix}$$

```
fsurf(f(x,y),[-1,1,-1.5,-0.2])
hold on
plot3(sol.x,sol.y,f(sol.x,sol.y),'.','MarkerSize',30)
hold off
```


0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.2
-1.4
-1.6
-0.4
-0.6
-0.8
-1
-1.2
-1.4
-1
0
1